

MSc Project Second Marker Meeting

Enge Xiong | Bilingual Preparation Notes / 中英对照会议准备稿

Meeting focus / 会议重点	Explain current progress, demonstrate technical understanding, discuss remaining work, and ask for feedback before final submission. 说明当前进度、展示技术理解、讨论剩余工作，并在最终提交前获取建议。
Project summary / 项目概述	A student support and mentor matching platform with registration, email verification, JWT authentication, user profiles, mentor matching, chat sessions, WebSocket messaging, and a voice-signalling module. 一个学生支持与导师匹配平台，包含注册、邮箱验证、JWT 认证、用户资料、导师匹配、会话聊天、WebSocket 实时消息以及语音信令模块。
Best opening / 开场一句	Thank you for meeting me. I will briefly explain what has been completed, what is still in progress, and where I would appreciate feedback. 感谢您抽时间见我。我会简要说明已经完成的部分、仍在推进的工作，以及希望获得建议的地方。

Use this as a speaking guide, not a script to read word-for-word. The English answers are intentionally concise so they sound natural in the meeting. / 这份材料适合当作口头表达提示，不需要逐字照读。英文回答刻意写得比较简洁，方便会议中自然表达。

1. Progress Snapshot / 当前进度概览

- Completed: registration and login, email verification, password reset, JWT-based authentication, profile management, mentor matching pool, session creation/closing, chat history, and WebSocket chat. / 已完成：注册登录、邮箱验证、密码重置、JWT 认证、个人资料管理、导师匹配池、会话创建与关闭、聊天记录和 WebSocket 实时聊天。
- Recent work: user profiles now include institution, programme and location; the matching logic uses status, problem type, institution, location and programme similarity. / 近期工作：用户资料增加学校、专业和地区；匹配逻辑结合状态、问题类型、学校、地区和专业相似度。
- In progress: voice communication through WebSocket signalling for WebRTC-style offer, answer and ICE candidate messages; integration and testing still need polishing. / 进行中：通过 WebSocket 转发 WebRTC 风格的 offer、answer 和 ICE candidate 语音信令；集成和测试还需要完善。
- Next: evaluation design, user testing, results analysis, final dissertation writing, and improving robustness around concurrent WebSocket sessions. / 下一步：设计评估、进行用户测试、分析结果、完成论文，并提升并发 WebSocket 会话的稳定性。

Technical points to mention if asked / 如被追问可提到的技术点

- The backend is built around FastAPI routers, SQLAlchemy models, Pydantic schemas, and separate service managers for text chat and voice signalling. / 后端以 FastAPI 路由、SQLAlchemy 数据模型、Pydantic schema，以及独立的聊天和语音连接管理服务为核心。
- The matching strategy first filters eligible mentors, then ranks candidates using weighted profile attributes. / 匹配策略先筛选可用导师，再基于用户资料属性进行加权评分。
- A known area to validate carefully is the voice module integration and edge cases such as expired tokens, closed sessions, and disconnected sockets. / 需要重点验证的部分是语音模块集成，以及 token 过期、会话关闭、socket 断开等边界情况。

2. Likely Questions / 可能被问到的问题

Q1. How is your project going?

中文回答

项目整体进展比较顺利。核心功能已经实现，包括注册登录、JWT 认证、邮箱验证、个人资料、导师匹配、会话管理和实时聊天。目前我主要在完善语音通信、测试系统稳定性，并准备评估和论文后续章节。

English answer

The project is progressing well. The core functionality is implemented, including registration and login, JWT authentication, email verification, user profiles, mentor matching, session management, and real-time chat. I am now refining voice communication, testing system stability, and preparing the evaluation and dissertation chapters.

Q2. What have you completed since the interim report?

中文回答

中期报告之后，我扩展了用户资料字段，加入学校、专业 and 地区信息；同时调整了匹配逻辑，让它不只是按用户类型和问题类型匹配，也能参考学校、地区和专业相似度。我还完善了会话和聊天相关接口，并开始加入语音信令模块。

English answer

Since the interim report, I expanded the user profile with institution, programme, and location information. I also improved the matching logic so it considers profile similarity, not only status and problem type. In addition, I refined the session and chat APIs and started adding the voice signalling module.

Q3. What are you working on now?

中文回答

我现在主要做三件事：第一是完成语音通信模块的集成和测试；第二是设计评估方法，包括功能测试和用户反馈；第三是继续整理论文结构，把实现、评估和讨论部分写完整。

English answer

I am currently focusing on three things: integrating and testing the voice communication module, designing the evaluation method including functional testing and user feedback, and continuing the dissertation chapters on implementation, evaluation, and discussion.

Q4. Can you explain your matching algorithm?

中文回答

匹配分两步。第一步是硬筛选，只选择状态和问题类型相同的可用导师。第二步是加权评分，例如同一学校给更高权重，同一地区给较低权重，专业相似度通过标准化后的文本相似度计算。最后选择得分最高的导师创建会话。

English answer

The matching process has two stages. First, it filters available mentors by status and problem type. Then it ranks candidates using weighted attributes: institution has a higher weight, location has a smaller weight, and programme similarity is calculated from normalized programme text. The highest-scoring mentor is selected for the session.

Q5. How will you evaluate the system?

中文回答

我计划从两个层面评估。技术层面会测试核心流程，例如注册、登录、匹配、聊天、语音信令和异常情况。用户层面会让少量测试用户完成典型任务，并收集可用性反馈，例如流程是否清晰、匹配是否合理、实时聊天是否稳定。

English answer

I plan to evaluate it on two levels. Technically, I will test the core flows such as registration, login, matching, chat, voice signalling, and error handling. From the user perspective, I will ask a small number of test users to complete typical tasks and collect usability feedback on clarity, matching quality, and real-time communication stability.

Q6. What difficulties have you encountered?

中文回答

主要困难是 WebSocket 会话管理和多用户同步，尤其是用户断开、会话关闭、token 过期时如何保持状态一致。另一个挑战是语音通信不仅涉及后端转发信令，还需要前端和浏览器端配合，所以集成测试比较重要。

English answer

The main challenge has been WebSocket session management and synchronisation between users, especially when a user disconnects, a session closes, or a token expires. Another challenge is that voice communication depends on both backend signalling and browser-side behaviour, so integration testing is important.

Q7. What still needs to be done?

中文回答

剩余工作包括完成语音模块测试、修复边界情况、完成用户测试和评估分析，并把论文的实现、评估、局限性和未来工作部分整理好。

English answer

The remaining work is to complete testing for the voice module, fix edge cases, conduct user testing, analyse the evaluation results, and finalise the implementation, evaluation, limitations, and future work sections of the dissertation.

Q8. Do you think you will finish on time?

中文回答

我认为可以按时完成。核心系统已经搭好，剩余工作主要是完善、测试、评估和写作。我会把语音功能作为优先事项，同时确保即使某些高级功能需要简化，核心系统和评估也能完整交付。

English answer

Yes, I believe I can finish on time. The core system is already in place, and the remaining tasks are mainly refinement, testing, evaluation, and writing. I will prioritise the voice feature, while making sure the core system and evaluation can still be delivered properly even if some advanced parts need to be simplified.

3. Strong Short Answers / 可快速使用的简短回答

Topic / 主题	Chinese / 中文	English / 英文
Main contribution 主要贡献	把学生和合适的导师连接起来，并支持实时沟通。	The main contribution is connecting students with suitable mentors and supporting real-time communication.
Technical stack 技术栈	FastAPI、SQLAlchemy、JWT、Pydantic、WebSocket，以及前端配合完成实时交互。	The system uses FastAPI, SQLAlchemy, JWT, Pydantic, WebSocket, and frontend integration for real-time interaction.
Why matching matters 为什么匹配重要	更相关的导师可以提高支持质量，也能减少学生等待和试错。	Better matching can improve support quality and reduce waiting time and trial-and-error for students.
Main risk 主要风险	语音模块和并发连接需要更多测试，所以我会优先验证这些部分。	The main risk is the voice module and concurrent connections, so I will prioritise testing those parts.
Evaluation evidence 评估证据	我会结合功能测试、典型任务测试和用户反馈来证明系统有效。	I will combine functional testing, task-based testing, and user feedback to demonstrate effectiveness.

4. Questions To Ask / 可以问老师的问题

- Is my current scope appropriate for the final MSc project submission? / 我目前的项目范围是否适合最终 MSc 项目提交？
- Should I prioritise deeper evaluation, more robust voice communication, or improving the matching algorithm? / 接下来我应该优先加强评估、语音通信稳定性，还是匹配算法？
- For the dissertation, how much implementation detail should I include in the main text versus appendix? / 论文中实现细节应该在正文写到什么程度，哪些适合放附录？
- What kind of evidence would make the evaluation section more convincing? / 哪类证据会让评估章节更有说服力？
- Are there any risks you think I should address before final submission? / 最终提交前，您认为还有哪些风险需要我优先处理？

Closing line / 结束语

中文回答 感谢您的建议。我会根据今天的反馈调整接下来的优先级，尤其是评估设计、语音模块测试和论文写作。	English answer Thank you for the feedback. I will use today's advice to adjust my priorities, especially around the evaluation design, voice-module testing, and dissertation writing.
--	---